# Robust Deepfake Detection Under Real-World Perturbations

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***Abstract*—This paper studies robust detection of manipulated face videos with a lightweight network. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.**

***Index Terms*—deepfake detection, forensics, robustness**

## I. Introduction

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

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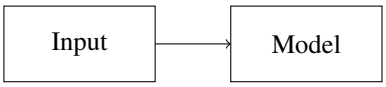



Fig. 1. Overview of the proposed pipeline.

TABLE I
Comparison with prior methods

| Method | Acc | AUC |
|---|---|---|
| hline Baseline | 81.2 | 0.85 Ours |
| 93.7 | 0.96 | |
| hline | | |

## II. Method

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. The loss is

$$L = \sum_{i} (y_i - \hat{y}_i)^2 \tag{1}$$

and the score is

$$s = \sigma(w^T x + b). \tag{2}$$

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## III. Experiments

Deepfake detection has become an important problem in multimedia forensics because modern generators produce

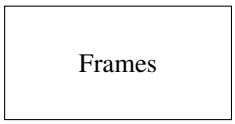



Fig. 2. Example frames.

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## IV. Conclusion

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

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